# Chapter 6: Application of Derivatives

## Practice Set A: step-wise solutions

**20** questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

## Level 1

8 marks

**1** MCQ Board-pattern — 1 mark

The radius of a circular plate is increasing at the rate of $0.2$ cm/s. The rate at which its area is increasing when the radius is $5$ cm is:

**(A)** $\pi \text{ cm}^2/\text{s}$ **(B)** $2\pi \text{ cm}^2/\text{s}$

**(C)** $10\pi \text{ cm}^2/\text{s}$ **(D)** $25\pi \text{ cm}^2/\text{s}$

**Formula / idea:** Chain rule: $\dfrac{dA}{dt} = \dfrac{dA}{dr} \cdot \dfrac{dr}{dt}$

1 Area of a circle
$A = \pi r^2$

2 Differentiate both sides w.r.t. $t$
$\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$

3 Put $r = 5$ and $\dfrac{dr}{dt} = 0.2$
$\dfrac{dA}{dt} = 2\pi(5)(0.2) = 2\pi \text{ cm}^2/\text{s}$ ✓ 1

**Answer** **(B)** $2\pi \text{ cm}^2/\text{s}$

**Watch out:** Forgetting to multiply by $\dfrac{dr}{dt}$ gives $10\pi$.

**2** MCQ CBSE SQP 2024-25 1 mark

The largest interval in which the function $f$ defined by $f(x) = e^x$ is strictly increasing, is:

**(A)** $[1, \infty)$ **(B)** $(-\infty, 0)$

**(C)** $(-\infty, \infty)$ **(D)** $(0, \infty)$

**Formula / idea:** If $f'(x) > 0$ for all $x$ in an interval, $f$ is strictly increasing there.

1 Differentiate
$f'(x) = e^x$

2 $e^x > 0$ for every real $x$
$f'(x) > 0$ for all $x \in \mathbb{R}$

3 So $f$ is strictly increasing on the whole real line ✓ 1
Largest interval $= (-\infty, \infty)$

**Answer** **(C)** $(-\infty, \infty)$

**Watch out:** Picking (A), (B) or (D): $f$ increases there too, but each is only a part of $\mathbb{R}$.

**3** MCQ CBSE SQP 2025-26 1 mark

A function $f(x) = 10 - x - 2x^2$ is increasing on the interval:

**(A)** $\left(-\infty, -\frac{1}{4}\right]$

**(B)** $\left(-\infty, \frac{1}{4}\right)$

**(C)** $\left[-\frac{1}{4}, \infty\right)$

**(D)** $\left[-\frac{1}{4}, \frac{1}{4}\right]$

**Formula / idea:** $f$ is increasing where $f'(x) \geq 0$.

1 Differentiate
$f'(x) = -1 - 4x$

2 Put $f'(x) \geq 0$
$-1 - 4x \geq 0 \Rightarrow 4x \leq -1$

3 Divide by $4$

$x \le -\frac{1}{4} \Rightarrow \left(-\infty, -\frac{1}{4}\right]$ ✓ 1

**Answer (A)** $\left(-\infty, -\frac{1}{4}\right]$

**Watch out:** Getting $x \ge -\frac{1}{4}$ by a sign slip, which gives (C), where $f$ actually decreases.

**4** MCQ CBSE 2025 1 mark

The absolute maximum value of the function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is:

**(A)** $0$ **(B)** $2$

**(C)** $4$ **(D)** $5$

**Formula / idea:** On $[a, b]$, compare $f$ at the critical points and at the end points.

1 Differentiate and equate to zero

$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

2 Only $x = 1$ lies in $[0, 2]$; find the values

$f(0) = 2, \quad f(1) = 0, \quad f(2) = 4$

3 Largest value

$\text{Absolute maximum} = 4 \text{ at } x = 2$ ✓ 1

**Answer (C)** $4$

**Watch out:** Checking only the critical point $x = 1$ and forgetting the end points.

**5** MCQ Board-pattern 1 mark

The local minimum value of $f(x) = x^2 + \frac{16}{x}$, $x > 0$, is:

**(A)** $2$ **(B)** $8$

**(C)** $12$ **(D)** $16$

**Formula / idea:** Second derivative test: $f'(c) = 0$ and $f''(c) > 0$ gives a local minimum at $c$.

1 Differentiate

$f'(x) = 2x - \frac{16}{x^2}$

2 Put $f'(x) = 0$

$2x^3 = 16 \Rightarrow x^3 = 8 \Rightarrow x = 2$

3 Second derivative test
$f''(x) = 2 + \frac{32}{x^3}, \quad f''(2) = 6 > 0$

4 So local minimum at $x = 2$; its value is
$f(2) = 4 + 8 = 12$ ✓ 1

**Answer** **(C)** 12

**Watch out:** Giving the point $x = 2$ (option A) instead of the value $f(2) = 12$.

**6** **Assertion-Reason** CBSE SQP 2023-24 1 mark

Let $f(x)$ be a polynomial function of degree 6 such that $\frac{d}{dx}\big(f(x)\big) = (x-1)^3(x-3)^2$. **Assertion (A):** $f(x)$ has a minimum at $x = 1$. **Reason (R):** When $\frac{d}{dx}\big(f(x)\big) < 0$ for all $x \in (a-h, a)$ and $\frac{d}{dx}\big(f(x)\big) > 0$ for all $x \in (a, a+h)$, where $h$ is an infinitesimally small positive quantity, then $f(x)$ has a minimum at $x = a$, provided $f(x)$ is continuous at $x = a$.

**(A)** **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

**(C)** Assertion (A) is true, but Reason (R) is false.

**(D)** Assertion (A) is false, but Reason (R) is true.

**Formula / idea:** First derivative test: $f'$ changes from $-$ to $+$ at $c$ means local minimum at $c$.

1 Near $x = 1$, $(x-3)^2 > 0$ and $(x-1)^3$ changes sign from $-$ to $+$
$x < 1 : f'(x) < 0, \qquad x > 1 : f'(x) > 0$

2 So $f$ has a minimum at $x = 1$: **A is true**

3 R is the first derivative test: **R is true**

4 A is obtained by applying R at $a = 1$: R explains A
Option (A) ✓ 1

**Answer** **(A) Both true; R is the correct explanation of A.**

**Watch out:** Thinking $(x-1)^3$ keeps one sign; an odd power changes sign at $x = 1$.

**7** **Short answer** CBSE 2025 2 marks

The side of an equilateral triangle is increasing at the rate of $3$ cm/s. At what rate is its area increasing when the side of the triangle is $15$ cm?

**Formula / idea:** Area of an equilateral triangle of side $x$ is $\frac{\sqrt{3}}{4}x^2$.

| | | |
|---|---|---|
| 1 | Let side $= x$; write the area $A = \frac{\sqrt{3}}{4}x^2$ | ✓ ½ |
| 2 | Differentiate w.r.t. $t$ $\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2x \frac{dx}{dt}$ | ✓ ½ |
| 3 | Put $x = 15$, $\frac{dx}{dt} = 3$ $\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2(15)(3) = \frac{45\sqrt{3}}{2}$ | ✓ ½ |
| 4 | Hence the area increases at $\frac{45\sqrt{3}}{2}\ \text{cm}^2/\text{s}$ | ✓ ½ |

**Answer** **Area increases at** $\frac{45\sqrt{3}}{2}\ \text{cm}^2/\text{s}$

**Watch out:** Dropping the $\frac{dx}{dt} = 3$ factor, or writing the units as cm/s.

## Level 2

19 marks

**8** **Short answer** CBSE 2025 3 marks

Find the intervals in which the function $f(x) = 5x^3 - 3x^2$ is (i) increasing (ii) decreasing.

**Formula / idea:** $f'(x) \geq 0$ gives increasing; $f'(x) \leq 0$ gives decreasing.

| | | |
|---|---|---|
| 1 | Differentiate and factorise $f'(x) = 15x^2 - 6x = 3x(5x - 2)$ | ✓ 1 |
| 2 | Put $f'(x) = 0$ $x = 0, \quad x = \frac{2}{5}$ | ✓ ½ |
| 3 | Check the sign of $f'(x)$ in each interval $x < 0 : f'(x) > 0$ $0 < x < \frac{2}{5} : f'(x) < 0$ $x > \frac{2}{5} : f'(x) > 0$ | ✓ ½ |

4 (i) $f$ is increasing in
$(-\infty, 0] \cup \left[\frac{2}{5}, \infty\right)$ ✓ ½

5 (ii) $f$ is decreasing in
$\left[0, \frac{2}{5}\right]$ ✓ ½

**Answer** **(i) Increasing in** $(-\infty, 0] \cup \left[\frac{2}{5}, \infty\right)$ **(ii) Decreasing in** $\left[0, \frac{2}{5}\right]$

**Watch out:** Skipping the sign check: $f'(x) < 0$ between $0$ and $\frac{2}{5}$, so $f$ falls there.

**9** **Short answer** CBSE SQP 2025-26 2 marks

A spherical ball of ice melts in such a way that the rate at which its volume decreases at any instant is directly proportional to its surface area. Prove that the radius of the ice ball decreases at a constant rate.

**Formula / idea:** $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$, and $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

1 Let radius $= r$ at time $t$
$V = \frac{4}{3}\pi r^3, \qquad S = 4\pi r^2$

2 Given: volume decreases at a rate proportional to $S$ $(k > 0)$
$\frac{dV}{dt} = -kS = -4k\pi r^2$ ✓ ½

3 Differentiate $V$ w.r.t. $t$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$ ✓ ½

4 Equate the two expressions
$4\pi r^2 \frac{dr}{dt} = -4k\pi r^2$ ✓ ½

5 Cancel $4\pi r^2$ on both sides
$\frac{dr}{dt} = -k$ (constant) ✓ ½

6 Hence the radius decreases at the constant rate $k$

**Answer** $\frac{dr}{dt} = -k$**, a constant; so the radius decreases at a constant rate.**

**Watch out:** Leaving out the minus sign: the volume is decreasing, so $\frac{dV}{dt}$ is negative.

**10** **Short answer** PYQ-style 3 marks

Find the local maximum and local minimum values of $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$.

**Formula / idea:** Second derivative test: $f''(c) < 0$ gives local max, $f''(c) > 0$ gives local min.

| | | |
|---|---|---|
| 1 | Differentiate and factorise<br>$f'(x) = 12x^3 + 12x^2 - 24x = 12x(x+2)(x-1)$ | ✓ ½ |
| 2 | Put $f'(x) = 0$<br>$x = -2,\ 0,\ 1$ | ✓ ½ |
| 3 | Second derivative<br>$f''(x) = 36x^2 + 24x - 24$ | ✓ ½ |
| 4 | At $x = -2$: $f''(-2) = 72 > 0$, local minimum<br>$f(-2) = 48 - 32 - 48 + 12 = -20$ | ✓ ½ |
| 5 | At $x = 0$: $f''(0) = -24 < 0$, local maximum<br>$f(0) = 12$ | ✓ ½ |
| 6 | At $x = 1$: $f''(1) = 36 > 0$, local minimum<br>$f(1) = 3 + 4 - 12 + 12 = 7$ | ✓ ½ |
| 7 | Hence local maximum value $= 12$; local minimum values $= -20$ and $7$ | |

**Answer** **Local maximum value $12$ (at $x = 0$); local minimum values $-20$ (at $x = -2$) and $7$ (at $x = 1$)**

**Watch out:** Missing the root $x = 0$ after taking $12x$ common.

**11** **Short answer** CBSE SQP 2023-24 3 marks

Find the interval/s in which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = xe^x$ is increasing.

**Formula / idea:** Product rule: $(uv)' = u'v + uv'$; $f$ increases where $f'(x) \geq 0$.

| | | |
|---|---|---|
| 1 | Differentiate by the product rule<br>$f'(x) = e^x + xe^x = e^x(x+1)$ | ✓ 1 |
| 2 | Put $f'(x) = 0$; $e^x$ is never zero<br>$x + 1 = 0 \ \Rightarrow \ x = -1$ | ✓ ½ |
| 3 | Sign of $f'(x)$, using $e^x > 0$<br>$x > -1 : f'(x) > 0$<br>$x < -1 : f'(x) < 0$ | ✓ 1 |

| | | |
|---|---|---|
| 4 | Hence $f$ is increasing in $[-1, \infty)$ | ✓ ½ |

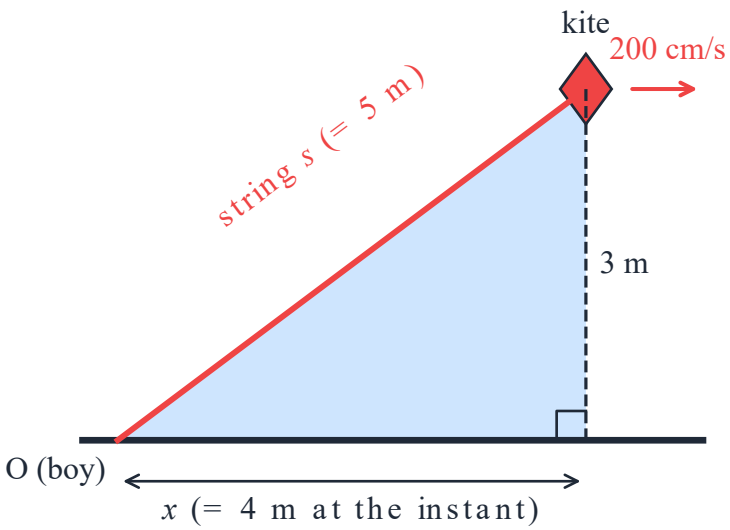
**Answer** $f$ **is increasing in** $[-1, \infty)$

**Watch out:** Trying to solve $e^x = 0$; it has no solution, so only $x + 1$ decides the sign.

**12** **Short answer** CBSE SQP 2024-25 3 marks

A kite is flying at a height of 3 metres and 5 metres of string is out. If the kite is moving away horizontally at the rate of 200 cm/s, find the rate at which the string is being released.

**Formula / idea:** Pythagoras theorem, then differentiate w.r.t. $t$.




| | | |
|---|---|---|
| 1 | Let horizontal distance $= x$ m and string $= s$ m; height is fixed at 3 m<br>$s^2 = x^2 + 3^2$ | ✓ ½ |
| 2 | When $s = 5$<br>$x = \sqrt{25 - 9} = 4$ m | ✓ ½ |
| 3 | Differentiate w.r.t. $t$<br>$2s\,\frac{ds}{dt} = 2x\,\frac{dx}{dt}$ | ✓ 1 |
| 4 | Put $x = 4$, $s = 5$, $\frac{dx}{dt} = 200$ cm/s $= 2$ m/s<br>$\frac{ds}{dt} = \frac{4 \times 2}{5} = 1.6$ m/s | ✓ ½ |
| 5 | Hence the string is released at 1.6 m/s, i.e. 160 cm/s | ✓ ½ |

**Answer** **String is released at** $1.6$ **m/s (**$= 160$ **cm/s)**

**Watch out:** Mixing cm/s with metres; convert 200 cm/s to 2 m/s first.

**13** **Short answer** CBSE 2025 3 marks

Find the absolute maximum and absolute minimum of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.

**Formula / idea:** On $[a, b]$, compare $f$ at all critical points and at both end points.

| | | |
|---|---|---|
| 1 | Differentiate and factorise<br>$f'(x) = 6x^2 - 30x + 36 = 6(x - 2)(x - 3)$ | ✓ ½ |
| 2 | Critical points (both lie in $[1, 5]$)<br>$x = 2, \quad x = 3$ | ✓ ½ |
| 3 | Values at critical points and end points<br>$f(1) = 24, \quad f(2) = 29$<br>$f(3) = 28, \quad f(5) = 56$ | ✓ 1 |
| 4 | Largest value<br>$\text{Absolute maximum} = 56 \text{ at } x = 5$ | ✓ ½ |
| 5 | Smallest value<br>$\text{Absolute minimum} = 24 \text{ at } x = 1$ | ✓ ½ |

**Answer** **Absolute maximum** $= 56$ **at** $x = 5$**; absolute minimum** $= 24$ **at** $x = 1$

**Watch out:** Taking the local max $f(2) = 29$ as the absolute maximum without checking $f(5)$.

**14** **Short answer** CBSE SQP 2023-24 2 marks

Find the maximum profit that a company can make, if the profit function is given by $P(x) = 72 + 42x - x^2$, where $x$ is the number of units and $P$ is the profit in rupees.

**Formula / idea:** $P'(x) = 0$ and $P''(x) < 0$ give the maximum.

| | | |
|---|---|---|
| 1 | Differentiate<br>$P'(x) = 42 - 2x$ | ✓ ½ |
| 2 | Put $P'(x) = 0$<br>$x = 21$ | ✓ ½ |
| 3 | Second derivative test<br>$P''(x) = -2 < 0 \Rightarrow \text{maximum at } x = 21$ | ✓ ½ |

| | | |
|---|---|---|
| 4 | Maximum profit<br>$P(21) = 72 + 882 - 441 = ₹\,513$ | ✓ ½ |

**Answer** **Maximum profit** $= ₹\,513$ **(at** $x = 21$ **units)**

**Watch out:** Stopping at $x = 21$; the question asks for the profit $P(21)$.

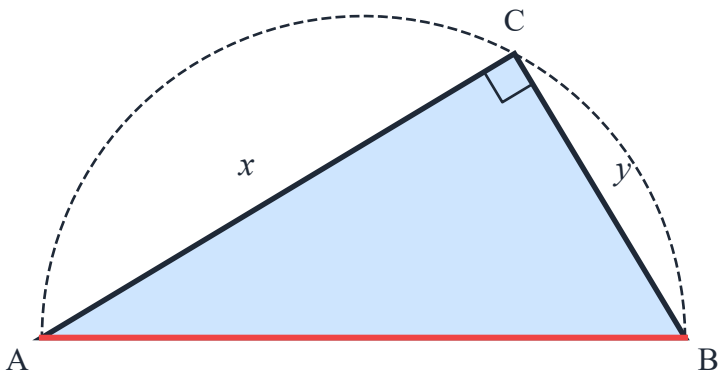
## Level 3

25 marks

15 **Long answer** Board-pattern 5 marks

Show that among all right-angled triangles with a given hypotenuse $h$, the isosceles triangle has the maximum area. Also find this maximum area.

**Formula / idea:** Area and (area)$^2$ are maximum together, so maximise $S = A^2$ to avoid the square root.




| | | |
|---|---|---|
| 1 | Let the legs be $x$ and $y$; by Pythagoras<br>$x^2 + y^2 = h^2 \;\Rightarrow\; y = \sqrt{h^2 - x^2}$ | ✓ 1 |
| 2 | Area of the triangle<br>$A = \frac{1}{2}x\sqrt{h^2 - x^2}$ | ✓ ½ |
| 3 | Let $S = A^2$<br>$S = \frac{1}{4}x^2(h^2 - x^2) = \frac{1}{4}(h^2x^2 - x^4)$<br>$A > 0$, so $A$ and $A^2$ are maximum together | ✓ ½ |
| 4 | Differentiate w.r.t. $x$<br>$\frac{dS}{dx} = \frac{1}{4}(2h^2x - 4x^3)$ | ✓ ½ |

5 Put $\dfrac{dS}{dx} = 0$ $(x \neq 0)$

$x^2 = \dfrac{h^2}{2} \Rightarrow x = \dfrac{h}{\sqrt{2}}$ ✓ ½

6 Second derivative test

$\dfrac{d^2S}{dx^2} = \dfrac{1}{4}(2h^2 - 12x^2) = \dfrac{1}{4}(2h^2 - 6h^2) = -h^2 < 0$ ✓ 1

7 So the area is maximum; find the other leg

$y = \sqrt{h^2 - \dfrac{h^2}{2}} = \dfrac{h}{\sqrt{2}} = x$ ✓ ½

8 Legs are equal, so the triangle is isosceles; maximum area

$A = \dfrac{1}{2} \cdot \dfrac{h}{\sqrt{2}} \cdot \dfrac{h}{\sqrt{2}} = \dfrac{h^2}{4}$ ✓ ½

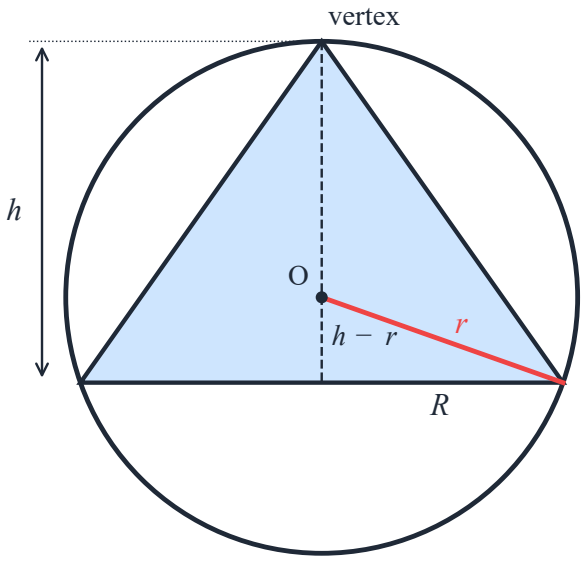
**Answer** **Area is maximum when both legs equal $\dfrac{h}{\sqrt{2}}$ (isosceles); maximum area $= \dfrac{h^2}{4}$**

**Watch out:** Differentiating $A$ directly and making an error with the square root; work with $S = A^2$.

16 **Long answer** PYQ-style 5 marks

Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius $r$ is $\dfrac{4r}{3}$. Also show that the maximum volume of the cone is $\dfrac{8}{27}$ of the volume of the sphere.

**Formula / idea:** Write $V$ in one variable $h$, then use $V'(h) = 0$ and $V''(h) < 0$.



sphere of radius $r$ (cross-section)

1 Cone: height $h$, base radius $R$; right triangle with hypotenuse $r$ gives

$R^2 + (h - r)^2 = r^2 \Rightarrow R^2 = 2rh - h^2$ ✓ 1

2 Volume of the cone

$$V = \frac{1}{3}\pi R^2 h = \frac{\pi}{3}(2rh^2 - h^3)$$

✓ ½

3 Differentiate w.r.t. $h$

$$\frac{dV}{dh} = \frac{\pi}{3}(4rh - 3h^2)$$

✓ ½

4 Put $\frac{dV}{dh} = 0$ $(h \neq 0)$

$$4r = 3h \;\Rightarrow\; h = \frac{4r}{3}$$

✓ ½

5 Second derivative at $h = \frac{4r}{3}$

$$\frac{d^2V}{dh^2} = \frac{\pi}{3}(4r - 6h) = \frac{\pi}{3}(4r - 8r) = -\frac{4\pi r}{3} < 0$$

✓ 1

6 So $V$ is maximum at altitude $h = \frac{4r}{3}$; then

$$R^2 = 2r \cdot \frac{4r}{3} - \frac{16r^2}{9} = \frac{8r^2}{9}$$

✓ ½

7 Maximum volume

$$V = \frac{1}{3}\pi \cdot \frac{8r^2}{9} \cdot \frac{4r}{3} = \frac{32\pi r^3}{81}$$

✓ ½

8 Compare with the sphere's volume $\frac{4}{3}\pi r^3$

$$\frac{32\pi r^3/81}{4\pi r^3/3} = \frac{32}{81} \cdot \frac{3}{4} = \frac{8}{27}$$

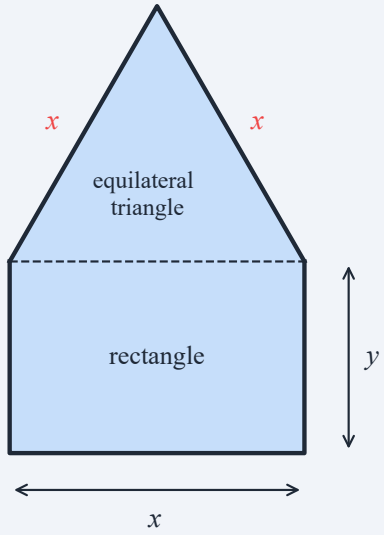
✓ ½

**Answer** **Altitude** $= \frac{4r}{3}$**; maximum volume** $= \frac{32\pi r^3}{81} = \frac{8}{27} \times \frac{4}{3}\pi r^3$

**Watch out:** Taking the distance from the centre to the base as $h$ instead of $h - r$.

17 **Long answer** PYQ-style 5 marks

A window is in the form of a rectangle surmounted by an equilateral triangle. The perimeter of the window is 12 m. Find the dimensions of the rectangle that will produce the largest area of the window.




**Formula / idea:** Use the perimeter to write area in one variable, then $A'(x) = 0$, $A''(x) < 0$.

| | | |
|---|---|---|
| 1 | Let width $= x$, height $= y$. Perimeter = base + 2 heights + 2 triangle sides<br>$x + 2y + 2x = 12 \Rightarrow y = \dfrac{12 - 3x}{2}$ | ✓ 1 |
| 2 | Area of window = rectangle + equilateral triangle<br>$A = xy + \dfrac{\sqrt{3}}{4}x^2$ | ✓ ½ |
| 3 | Substitute $y$<br>$A = 6x - \dfrac{3}{2}x^2 + \dfrac{\sqrt{3}}{4}x^2$ | ✓ ½ |
| 4 | Differentiate w.r.t. $x$<br>$\dfrac{dA}{dx} = 6 - 3x + \dfrac{\sqrt{3}}{2}x$ | ✓ ½ |
| 5 | Put $\dfrac{dA}{dx} = 0$<br>$x\left(3 - \dfrac{\sqrt{3}}{2}\right) = 6 \Rightarrow x = \dfrac{12}{6 - \sqrt{3}}$ | ✓ ½ |
| 6 | Second derivative test<br>$\dfrac{d^2A}{dx^2} = -3 + \dfrac{\sqrt{3}}{2} < 0 \Rightarrow \text{maximum}$ | ✓ 1 |
| 7 | Find $y$<br>$y = 6 - \dfrac{3}{2} \cdot \dfrac{12}{6 - \sqrt{3}} = \dfrac{18 - 6\sqrt{3}}{6 - \sqrt{3}}$ | ✓ ½ |

| | | |
|---|---|---|
| 8 | Hence, for the largest area, the rectangle has<br>$\text{width} = \dfrac{12}{6-\sqrt{3}}\ \text{m}, \quad \text{height} = \dfrac{18-6\sqrt{3}}{6-\sqrt{3}}\ \text{m}$ | ✓ ½ |

**Answer** **Width** $= \dfrac{12}{6-\sqrt{3}}$ **m** ($\approx 2.81$ **m**), **height** $= \dfrac{18-6\sqrt{3}}{6-\sqrt{3}}$ **m** ($\approx 1.78$ **m**)

**Watch out:** Counting the common side (top of the rectangle) in the perimeter: it is not on the boundary.

**18** **Short answer** PYQ-style 3 marks

Find the intervals in which the function $f(x) = \sin^4 x + \cos^4 x$, $0 \le x \le \dfrac{\pi}{2}$, is strictly increasing or strictly decreasing.

**Formula / idea:** $2\sin x\cos x = \sin 2x$, $\cos^2 x - \sin^2 x = \cos 2x$, $2\sin 2x\cos 2x = \sin 4x$

| | | |
|---|---|---|
| 1 | Differentiate<br>$f'(x) = 4\sin^3 x\cos x - 4\cos^3 x\sin x$ | ✓ ½ |
| 2 | Take $-4\sin x\cos x$ common and use double-angle formulas<br>$f'(x) = -4\sin x\cos x(\cos^2 x - \sin^2 x) = -2\sin 2x\cos 2x = -\sin 4x$ | ✓ ½ |
| 3 | Put $f'(x) = 0$ for $0 < x < \dfrac{\pi}{2}$, i.e. $0 < 4x < 2\pi$<br>$\sin 4x = 0 \Rightarrow 4x = \pi \Rightarrow x = \dfrac{\pi}{4}$ | ✓ ½ |
| 4 | For $0 < x < \dfrac{\pi}{4}$: $0 < 4x < \pi$, so $\sin 4x > 0$<br>$f'(x) < 0 \Rightarrow$ strictly decreasing in $\left(0, \dfrac{\pi}{4}\right)$ | ✓ 1 |
| 5 | For $\dfrac{\pi}{4} < x < \dfrac{\pi}{2}$: $\pi < 4x < 2\pi$, so $\sin 4x < 0$<br>$f'(x) > 0 \Rightarrow$ strictly increasing in $\left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$ | ✓ ½ |

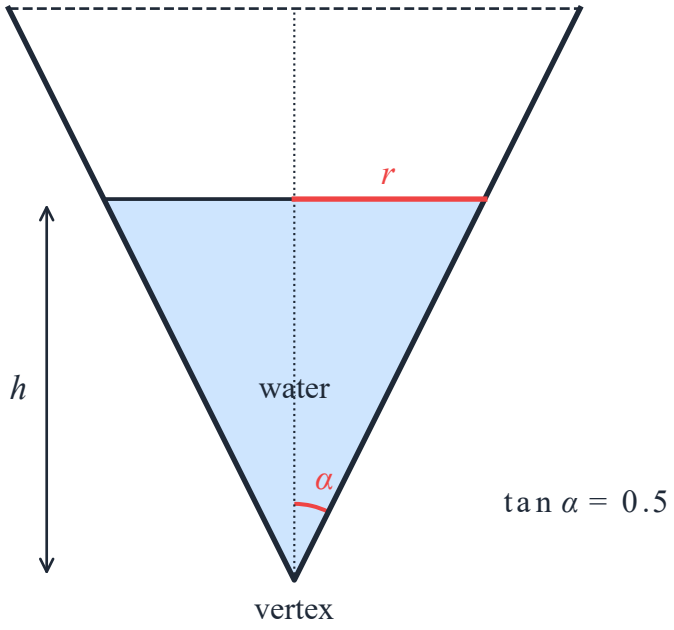
**Answer** **Strictly decreasing in** $\left(0, \dfrac{\pi}{4}\right)$**; strictly increasing in** $\left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$

**Watch out:** Forgetting the minus sign in $f'(x) = -\sin 4x$, which swaps the two answers.

19 **Short answer** PYQ-style 3 marks

A water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of $5\ \text{m}^3/\text{h}$. Find the rate at which the level of the water is rising at the instant when the depth of water in the tank is $4$ m.

**Formula / idea:** Similar triangles fix $r$ in terms of $h$; then $V = \dfrac{1}{3}\pi r^2 h$ in one variable.




| | | |
|---|---|---|
| 1 | Let depth $= h$ and radius of water surface $= r$; semi-vertical angle $\alpha$ has $\tan\alpha = 0.5$<br>$\dfrac{r}{h} = 0.5 \Rightarrow r = \dfrac{h}{2}$ | ✓ ½ |
| 2 | Volume of water in terms of $h$<br>$V = \dfrac{1}{3}\pi\left(\dfrac{h}{2}\right)^2 h = \dfrac{\pi h^3}{12}$ | ✓ ½ |
| 3 | Differentiate w.r.t. $t$<br>$\dfrac{dV}{dt} = \dfrac{\pi h^2}{4}\dfrac{dh}{dt}$ | ✓ 1 |
| 4 | Put $\dfrac{dV}{dt} = 5$, $h = 4$<br>$5 = \dfrac{\pi(16)}{4}\dfrac{dh}{dt} \Rightarrow \dfrac{dh}{dt} = \dfrac{5}{4\pi}$ | ✓ ½ |
| 5 | Hence the water level rises at<br>$\dfrac{5}{4\pi}\ \text{m/h}$ | ✓ ½ |

**Answer** **The water level rises at $\dfrac{5}{4\pi}$ m/h ($\approx 0.40$ m/h)**

**Watch out:** Differentiating $V = \dfrac{1}{3}\pi r^2 h$ with both $r$ and $h$ varying; replace $r$ by $\dfrac{h}{2}$ first.

**20** **Case study** CBSE SQP 2025-26 4 marks

**Case Study:** LED bulbs are energy-efficient because they use much less electricity than traditional bulbs while producing the same amount of light. A company manufactures a new type of energy-efficient LED bulb. The cost of production and the revenue generated by selling $x$ bulbs (in an hour) are modelled as $C(x) = 0.5x^2 - 10x + 150$ and $R(x) = -0.3x^2 + 20x$ respectively, where $C(x)$ and $R(x)$ are both in ₹. Use the given models to answer the following questions.

**(i)** Derive the profit function $P(x)$. 1 mark

**(ii)** Find the critical points of $P(x)$. 1 mark

**(iii)(a)** Determine whether the critical points correspond to a maximum or a minimum profit by using the second derivative test. 2 marks

OR

**(iii)(b)** Identify the possible practical value of $x$ (i.e. the number of bulbs that can realistically be produced and sold) that can maximise the profit, if the resources available and the expenditure on machines allow the company to produce a minimum of 10 but not more than 18 bulbs per hour. Also calculate the maximum profit. 2 marks

**Formula / idea:** Profit = Revenue − Cost; $P'(x) = 0$ gives critical points; $P''(x) < 0$ gives a maximum.

### (i) 1 mark

1 Profit = Revenue − Cost
$P(x) = R(x) - C(x) = (-0.3x^2 + 20x) - (0.5x^2 - 10x + 150)$ ✓ ½

2 Simplify
$P(x) = -0.8x^2 + 30x - 150$ ✓ ½

**Answer (i)** $P(x) = -0.8x^2 + 30x - 150$

### (ii) 1 mark

1 Differentiate
$P'(x) = -1.6x + 30$ ✓ ½

2 Put $P'(x) = 0$
$x = \dfrac{30}{1.6} = 18.75$ ✓ ½

**Answer (ii)** **Only critical point:** $x = 18.75$

### (iii)(a) 2 marks

1 Second derivative
$P''(x) = -1.6$ ✓ 1

2 At $x = 18.75$, $P'' < 0$. Hence the critical point gives maximum profit
$P''(18.75) = -1.6 < 0 \Rightarrow \text{maximum}$ ✓ 1

**Answer (iii)(a)** $P''(18.75) = -1.6 < 0$, **so** $x = 18.75$ **gives maximum profit**

OR

**(iii)(b)** 2 marks

| | | |
|---|---|---|
| 1 | For $x < 18.75$, $P'(x) > 0$, so $P$ is increasing on $[10, 18]$<br>$P'(x) = -1.6x + 30 > 0$ for $10 \leq x \leq 18$ | ✓ 1 |
| 2 | So profit is greatest at the largest allowed value<br>$x = 18$ bulbs per hour | ✓ ½ |
| 3 | Maximum profit<br>$P(18) = -0.8(324) + 540 - 150 = 130.8$ | ✓ ½ |

**Answer (iii)(b)** $x = 18$ **bulbs per hour; maximum profit** $= ₹\,130.80$

**Watch out:** In (iii)(b), choosing $x = 19$ (nearest whole number to $18.75$); only $10 \leq x \leq 18$ is allowed.